A · PATIENT DEMOGRAPHICS, VITALS & MEDICATION / SAFETY SCREEN

INTAKE · SAFETY

Patient Name _____ Age _____ Sex ☐ M ☐ F Distribution ☐ Central ☐ L ☐ R ☐ Bilat Date _____

Occupation / Sport _____ Work ☐ Full duty ☐ Modified ☐ Off work Onset / Surgery _____ Vitals: BP _____ HR _____ SpO_2 _____ Temp _____

Presentation ☐ Axial low back pain only ☐ Somatic buttock / thigh referral (above knee) ☐ Unilateral radicular leg pain (below knee) ☐ Bilateral neurogenic claudication ☐ Paresthesia / numbness / motor loss

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 Best _____ Worst _____ Back vs Leg dominant _____

☐ EPA contraindications (Houghton 2010): implanted electronic devices, pregnancy, malignancy, recently irradiated tissue, active DVT, bleeding, impaired sensation — Level: Guideline

☐ Anticoagulants / coagulopathy before needling (Kearns 2019): caution — possible increased incidence or severity of haematoma and bruising [E]

B · EMERGENCY SCREENING & RED FLAGS — COMPLETE BEFORE ANY LOADING OR MANUAL THERAPY

TRIAGE GATE

CAUDA EQUINA SYNDROME (CES) — SAME-DAY EMERGENCY

Surgical Emergency

☐ **Cauda equina syndrome** (i) — activate **same-day emergency** pathway for MRI & neurosurgical / spinal review

☐ **Urinary retention:** Sn 0.90, Sp 0.99 (classic estimate, Deyo 1992 JAMA) [L]

☐ **Bladder scan rule: PVR > 200 mL → MRI-confirmed CES in 43%**; PVR < 200 mL without objective signs → MRI+ rare (Ann R Coll Surg Engl 2024) [L] • PVR: _____ mL

☐ **Timing & prognosis: decompression < 48 h associated with better outcomes** (Ahn 2000 meta-analysis) [L]; bladder status at the time of surgery is the dominant factor for bladder recovery (Qureshi & Sell 2007) [L]

☐ Saddle sensory loss ☐ Voiding difficulty / retention ☐ Faecal incontinence ☐ Bilateral sciatica / motor loss

VERTEBRAL FRACTURE RED FLAGS (POST-TEST PROBABILITY)

Downie 2013 BMJ

☐ **Fracture red flags** [M]: **older age 9% (3–25) • prolonged corticosteroid use 33% (10–67) • severe trauma 11% (8–16) • contusion/abrasion 62% (49–74) • multiple red flags 90% (34–99)**

☐ **Cochrane fracture rule:** most single red flags uninformative; use in combination (Han 2023 Cochrane) [M]

MALIGNANCY, VASCULAR, INFLAMMATORY & INFECTION

Systemic & NICE NG59

☐ **Malignancy red flag** (Downie 2013 BMJ) [M]: **history of cancer 7% (3–16) in primary care; 33% (22–46) in emergency care**

☐ **Abdominal aortic aneurysm** (AAA palpation, Fink 2000 Arch Intern Med) [M]: **Sn 68% (60–76), Sp 75% (68–82), LR+ 2.7, LR− 0.43; Sn 82% for AAA ≥ 5.0 cm; Sn 91% if girth < 100 cm vs 53% if ≥ 100 cm**

☐ **Axial spondyloarthritis** (ASAS criteria, Rudwaleit 2011) [M]: Sn 79.5%, Sp 83.3% • chronic inflammatory back pain → rheumatology referral

☐ **Spinal infection** [E]: fever/rigors, IV drug use, immunosuppression, recent spinal procedure, relentless non-mechanical pain → urgent medical workup

☐ **Imaging rule** NICE NG59: **"Do not routinely offer imaging in a non-specialist setting" (1.1.4)**; consider imaging in specialist settings **"only if the result is likely to change management" (1.1.6)** • ref: Jarvik & Deyo 2002

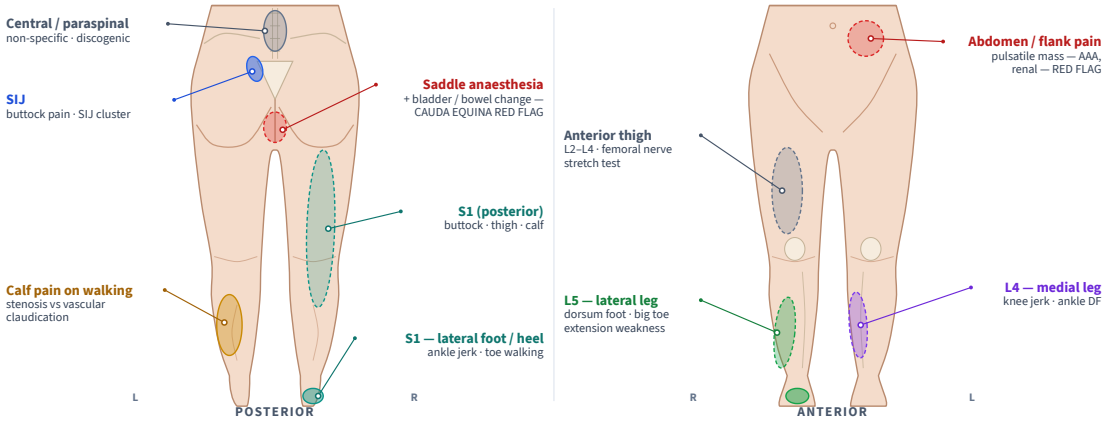
TRIAGE DISPOSITION: ☐ **Cleared — proceed with clinical dossier** ☐ Specialist / rheumatology referral ☐ Urgent medical / oncology workup ☐ **Same-day emergency MRI / ED transfer (CES / AAA)**

C · LUMBOSACRAL PAIN DISTRIBUTION MAP, PATHWAY ROUTER & SUBJECTIVE BEHAVIOUR

BODY CHART · SUBJECTIVE

PAIN MAP: shade symptom area · ☐ Shift / antalgic gait ☐ Centralises ☐ Peripheralises ☐ Heel / toe walk weak ☐ SLR / Slump (+)

Routes: Sciatica 3 4 8 Mechanical 2 3 7 8 Post-op 5 6 9



MECHANICAL AGGRAVATORS & EASING FACTORS

Behaviour

☐ Sitting / flexion / lifting (discogenic / radicular) ☐ Standing / walking / extension (stenosis / facet / pars)
☐ Cough / sneeze / Valsalva (dural / root) ☐ Sit-to-stand / rolling (instability / SIJ)
☐ Seated flexion relieves leg symptoms (claudication)

PATIENT GOALS — PSFS (0–10)

[L]

1. _____ /10 2. _____ /10 3. _____ /10

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

Differential Diagnosis, Risk Stratification & Screening

Differential & prognostic stratification: evaluate all 8 clinical patterns, administer risk stratification (STarT Back / ÖMPSQ-10), test repeated movements for centralisation / directional preference, and screen the hip, sacroiliac joint, and lower-limb neural integrity.

1 · NEURAL, CANAL & SOMATIC SPINE PATTERNS — MATRIX A

- ☐ **Discogenic / Radicular (Sciatica):** leg pain > back pain in dermatomal distribution; (+) SLR ▸ / crossed SLR ▸ / Slump ▸ or FST ▸ (p4); check centralisation [M]
- ☐ **Lumbar spinal stenosis cluster** (Cook 2011) [L]: bilateral symptoms, leg > back pain, pain during walking/standing, relief upon sitting, older age → **≥ 4 of 5 findings → post-test probability 76%**
- ☐ **Facet joint / somatic referred pain:** paraspinal tenderness, somatic referral into buttock/thigh without radicular deficit, aggravated by extension/rotation [E]
- ☐ **SIJ provocation cluster** (Laslett 2005) [M]: **≥ 3 of 6 tests: Sn 94%, Sp 78%; any 2 of the best 4 tests: AUC 0.842; all 6 negative → SIJ ruled out**

2 · INSTABILITY, PARS, INFLAMMATORY & VISCERAL/VASCULAR — MATRIX B

- ☐ **Lumbar segmental instability:** (+) PLET ▸, (+) PIT ▸, Fritz instability predictors (p3), aberrant motions, Hicks stabilisation CPR (p4) [L]
- ☐ **Spondylolysis — one-legged hyperextension test** (Masci 2006 BJSM) [L]: **"neither sensitive nor specific" for active spondylolysis** • see adolescent athlete RTS on p8
- ☐ **Axial spondyloarthritis (ASAS)** (Rudwaleit 2011) [M]: Sn 79.5%, Sp 83.3% • morning stiffness improving with exercise, not rest
- ☐ **Visceral / vascular referral:** AAA abdominal palpation (p1) [M], renal/ureteric, retroperitoneal, gastrointestinal or pelvic pathology (non-mechanical pain) [E]

3 · RISK STRATIFICATION (START BACK & ÖMPSQ-10) — STRATIFICATION

- ☐ **NICE on stratification** NICE NG59: **"Consider using risk stratification (for example, the STarT Back risk assessment tool) at first point of contact" (1.1.2)**
- ☐ **STarT Back — scoring** Calc (Hill 2008) [M]: **9 items; total 0–3 = low risk; psychosocial subscale (items 5–9) 4–5 = high risk; rest = medium**
- ☐ **STarT Back — stratified care (RCT, n = 851)** (Hill 2011 Lancet) [H]: **RMDQ between-group difference 1.81 (1.06–2.57) at 4 months; 1.06 (0.25–1.86) at 12 months**
- ☐ **STarT Back in older adults** (Vigdal 2024) [M]: **predictive validity poor; AUC 0.65–0.67**
- ☐ **Örebro short form (ÖMPSQ-10)** (Linton 2011) [M]: **correlation with long form 0.91; cut-off 50 identified 85% (occupational) / 83% (primary care) of those who developed disability**

STarT Back: Total _____ Psych _____ → ☐ Low ☐ Medium ☐ High
ÖMPSQ-10: _____ (☐ ≥ 50 risk)

4 · TBC, DIRECTIONAL PREFERENCE & WADDELL CAUTION — SUBGROUPING

- ☐ **Treatment-based classification (TBC)** (Fritz 2003) [M]: **TBC group: greater Oswestry change at 4 weeks (P = 0.023) and better return to full duty (P = 0.017) vs guideline-based care**
- ☐ **McKenzie directional preference (DP)** (Long 2004) [M]: **DP elicited in 74% (230) of subjects; exercises matching the DP improved every outcome vs opposite / non-directional**

Centralisation Status
☐ Centralises ☐ Peripheralises ☐ None

Directional Preference (DP)
☐ Ext ☐ Flex ☐ Lat shift ☐ No DP

- ☐ **Waddell signs** (Fishbain 2003 review) [M]: **associated with lower function, poorer non-surgical outcome and more pain; NOT evidence of secondary gain or psychological distress; cannot separate organic from non-organic**

5 · WORKING HYPOTHESIS & TBC SUBGROUP — SYNTHESIS

Primary Driver: _____
Secondary / Adjacent: _____
Psychosocial / Work: _____

Irritability: ☐ Low ☐ Moderate ☐ High

TBC Match: ☐ Specific Ex / DP ☐ Manipulation ☐ Stabilisation

6 · MANDATORY HIP, SACROILIAC JOINT & NEURAL SCREEN — CLEARING SCREEN

HIP & SACROILIAC CLEARING — Adjacent

- ☐ **Hip Joint Screen:** passive hip IR/ER & flexion, FADIR / FABER [E]: ☐ Cleared ☐ Hip positive
- ☐ **Laslett SIJ Cluster** [M]: ☐ Distraction ☐ Thigh thrust ☐ Compression ☐ Sacral thrust ☐ Gaenslen L ☐ Gaenslen R

STENOSIS & NEURAL SCREEN — Cluster

- ☐ **Cook Stenosis Items:** ☐ Bilat ☐ Leg>Back ☐ Walk/stand pain ☐ Sit relief ☐ Older age
- ☐ **Neural Screen (p3/p4):** ☐ Myotomes ☐ Dermatomes ☐ Reflexes ☐ SLR/Slump/FST

DIFFERENTIAL REASONING · STRATIFICATION PLAN · CLEARING SCREEN NOTES

1 · ACTIVE LUMBAR RANGE OF MOTION & REPEATED MOVEMENT ASSESSMENT

BLANK WRITE-IN FIELDS · NO NORMS

LUMBAR MOTION	BASELINE	FOLLOW-UP	SYMPTOM REPRODUCTION & END-FEEL	REPEATED MOVEMENT RESPONSE (DP)
Lumbar Flexion				☐ Centralises ☐ Peripheralises ☐ No change
Lumbar Extension				☐ Centralises ☐ Peripheralises ☐ No change
Left Lateral Flexion / Side-Glide				☐ Centralises ☐ Peripheralises ☐ No change
Right Lateral Flexion / Side-Glide				☐ Centralises ☐ Peripheralises ☐ No change
Thoracolumbar Rotation (L / R)				☐ Symmetric ☐ Painful / restricted

Modified-modified Schober (Tousignant 2005) [M]: intra ICC 0.95 (0.89–0.97); inter ICC 0.91 (0.83–0.96); minimum detectable change 1 cm; validity r = 0.67 • trained raters interrater ICC 0.942 (Johnson 2021) [L] • more reliable than double inclinometry; different metrics (MacDermid 2014) [L] • occiput-to-wall: reference value of 0 (Assassi 2014) [M] — MMS: ____ cm • OWD: ____ cm

2 · LOWER-LIMB NEUROLOGICAL EXAMINATION GRID (L2–S1)

DERMATOME · MYOTOME · REFLEX

ROOT	DERMATOME (LIGHT TOUCH / PINPRICK)	MYOTOME (KEY MUSCLE ACTION)	REFLEX	LEFT	RIGHT
L2	Anteromedial mid-thigh [E]	Hip flexion (Iliopsoas)	—		
L3	Distal anteromedial thigh & medial knee [E]	Knee extension (Quadriceps)	Patellar tendon (L3–L4)		
L4	Medial leg & medial malleolus [E]	Ankle dorsiflexion (Tibialis anterior / heel walk)	Patellar tendon (L4)		
L5	Anterolateral leg & dorsum of foot / first web space [E]	Great toe extension (EHL) & hip abduction	—		
S1	Posterior calf, lateral heel & lateral foot [E]	Ankle plantarflexion (Gastrocnemius-soleus / toe walk)	Achilles tendon (S1, p4)		
UMN / Saddle	Perineal / saddle sensation (CES screen p1) • Plantar response (Babinski) • Ankle clonus [E]			☐ Intact	☐ Abnormal

3 · SEGMENTAL MOBILITY TESTING & CLINICAL INSTABILITY BATTERY

RELIABILITY WARNING · PLET · PIT

- ☐ **Segmental mobility testing — reliability warning** (Hicks 2003) [L]: **κ −0.02 to 0.26 (not reliable); pain provocation κ 0.25–0.55** — rely on pain provocation rather than stiffness grading
- ☐ **Prone instability test (PIT) — reliability** [L]: **κ = 0.87 (Hicks 2003) BUT κ = 0.10 (−0.27 to 0.47) in a later study (Ravenna 2011) — reliability inconsistent**
- ☐ **Passive lumbar extension (PLE) test** (Kasai 2006; review confirming it as most accurate: Alqarni 2011) [L]: **Sn 84.2%, Sp 90.4%, LR+ 8.84 (4.51–17.33)**
- ☐ **Radiographic instability predictors** (Fritz 2005 Eur Spine J) [L]: **lumbar flexion ≥ 53° OR no hypomobility on segmental testing → LR+ 4.3 (1.8–10.6)**

PA Pain Provocation: ☐ L1 ☐ L2 ☐ L3 ☐ L4 ☐ L5 ☐ S1 **Instability Tests:** ☐ PLET (+) ☐ PIT (+) ☐ No hypomobility

4 · MCGILL ENDURANCE RATIOS & SØRENSEN

RATIOS ONLY · NO SECOND NORMS

- ☐ **McGill endurance battery** (McGill 1999) [L]: **reliability > 0.97; men side bridge = 65% of extensor time, 99% of flexor time; women 39% and 79% — ratios only, no second norms**
- ☐ **Biering-Sørensen** (Biering-Sørensen 1984) [L]: **good isometric back-extensor endurance may prevent first-time LBP in men — no second norms**

Flexor _____ Extensor _____ Side Bridge L _____ Side Bridge R _____

Flex / Ext Ratio _____ SB / Ext Ratio _____ SB L / R Ratio _____

5 · MOVEMENT QUALITY & ABERRANT MOVEMENTS

MOTOR CONTROL

- ☐ **Painful arc in flexion:** symptoms provoked in mid-range arc of forward bending [L]
- ☐ **Painful arc on return:** symptoms provoked during return from flexion to upright [L]
- ☐ **Instability catch / judder:** sudden acceleration, deceleration, or out-of-plane deviation [L]
- ☐ **Gower sign ("thigh climbing"):** hands walk up thighs to assist trunk extension [L]
- ☐ **Reversal of lumbopelvic rhythm:** knee flexion & anterior pelvic shift on return [L]

6 · HAND-HELD DYNAMOMETRY (HHD) STRENGTH & LIMB SYMMETRY

Calc

WRITE-IN · SAME DEVICE / POSITION

MOVEMENT	LEVER ARM (CM)	FORCE (N) INV / UNINV	TORQUE (N·M) = N × CM ÷ 100	LSI % = INV ÷ UNINV × 100
Hip abduction				
Hip extension				
Trunk extension				

Method [M]: belt-stabilised HHD vs isokinetic r = 0.60–0.90 but absolute values not concordant — the two devices cannot be interchanged (Martins 2017); hip peak torque ICC 0.848–0.899 (belt) and 0.848–0.942 (tension) (Batista 2024) • same device, same position, same strap / lever-arm point and same tester at every visit; compare with the other side [E]

OBJECTIVE EXAMINATION NOTES · PALPATION · MOTOR CONTROL OBSERVATIONS

Clinician _____ Date _____

How to read: copy & interpret diagnostic accuracy within its clinical context. Underline = Atlas card · ▸ = verified video · Calc = interactive scale calculator · Guide = clinical guideline.

1 · NEURODYNAMIC & NERVE ROOT IMPINGEMENT TESTS

SCIATIC · FEMORAL · REFLEX

☐	**Straight Leg Raise (SLR)** ▸ (Cochrane, mostly surgical samples — van der Windt 2010)	☐ L ☐ R	Sn 0.92 (0.87–0.95), Sp 0.28 (0.18–0.40) M
☐	**Crossed SLR** ▸ (well-leg raise reproducing concordant symptomatic leg pain — van der Windt 2010)	☐ L ☐ R	Sp 0.90 (0.85–0.94), Sn 0.28 (0.22–0.35) M
☐	**Slump vs SLR** ▸ (seated dural/sciatic tension with cervical differentiation — Majlesi 2008)	☐ L ☐ R	Slump Sn 0.84, Sp 0.83; SLR Sn 0.52, Sp 0.89 L
☐	**Femoral stretch test (prone knee bend)** ▸ (mid-lumbar impingement — Suri 2011 Spine)	☐ L ☐ R	FST + patellar reflex LR 7.0 (2.3–21); FST at L3 LR 5.7 L
☐	**Achilles reflex** (low-lumbar impingement — Suri 2011 Spine)	☐ L ☐ R	low-lumbar impingement LR 7.1 (0.96–53) L

2 · MANIPULATION CPR (FLYNN 2002 / CHILDS 2004)

5 CRITERIA CHECKLIST

Flynn 2002: 5 criteria; ≥ 4/5 → LR+ 24.38; probability of success 45% → 95% L
Childs 2004 validation: rule-positive + manipulation: OR 60.8 (5.2–704.7); 92% chance of success; NNT 1.9 (1.4–3.5) at 4 weeks M

- ☐ **Criterion:** Acute / recent onset of current low back pain episode
- ☐ **Criterion:** No symptoms extending distal to the knee
- ☐ **Criterion:** Low fear-avoidance beliefs on the FABQ-work subscale
- ☐ **Criterion:** Hypomobility on lumbar posteroanterior segmental testing
- ☐ **Criterion:** Adequate hip internal rotation range of motion

Criteria ticked: _____ / 5 ☐ ≥ 4/5 Rule-Positive (pair with exercise, p7)

3 · STABILISATION CPR (HICKS 2005)

KEY VARIABLES CHECKLIST

Hicks 2005: success LR+ 4.0; failure LR− 0.18; key variables: age, SLR, prone instability test, aberrant motions, hypermobility, fear-avoidance L

- ☐ **Key variable:** Age (younger age predictor)
- ☐ **Key variable:** Straight Leg Raise (SLR ▸) range of motion
- ☐ **Key variable:** Positive Prone Instability Test (PIT) ▸
- ☐ **Key variable:** Aberrant motions present during active flexion / return
- ☐ **Key variable:** Lumbar segmental hypermobility on PA spring testing
- ☐ **Key variable:** Fear-avoidance beliefs (FABQ)

☐ Success profile (LR+ 4.0) ☐ Failure profile (LR− 0.18)

4 · VALIDATED PATIENT-REPORTED OUTCOME MEASURES (PROMS DASHBOARD)

PROMS · VERIFIED MCIDS ONLY

MIC consensus (LBP — Ostelo 2008) M **:** VAS 15; NRS 2; RMDQ 5; ODI 10; or 30% improvement from baseline

INSTRUMENT & CALCULATOR	BASELINE	FOLLOW-UP	VERIFIED MCID / MIC & PSYCHOMETRIC PROPERTIES
ODI / Modified Oswestry Calc M			**ODI 10; or 30% improvement from baseline** (Ostelo 2008) • Modified Oswestry: **MCID ≈ 6 points** (Fritz 2001)
Roland-Morris (RMDQ, 24 yes/no) Calc M			**RMDQ 5; or 30% improvement from baseline** (Ostelo 2008) • **clinical improvement = 30% reduction from baseline + better on global rating** (Jordan 2006)
NPRS / VAS Pain Calc M			**MDC 2 points; MCID 2.2 (1 week) / 1.5 (4 weeks)** (Childs 2005) • MIC consensus: **VAS 15; NRS 2; or 30% improvement from baseline** (Ostelo 2008)
PSFS (chronic LBP) Calc L			**MCID 2 points (ROC); 1.4 (SEM method)** (Maughan 2010)
TSK-11 (11 items, 1–4) Calc M			**reduction ≥ 4 points = important change; ICC 0.81, SEM 2.54** (Woby 2005)
FABQ-work M			**predicts 4-week disability and work status in acute LBP** (Fritz 2001 Pain)
Pain Catastrophizing Scale (PCS) E			track against the patient's baseline
STarT Back (9 items) Calc M			**total 0–3 = low risk; psychosocial subscale (items 5–9) 4–5 = high risk; rest = medium** (Hill 2008)

NEURODYNAMIC SYNTHESIS · CPR SUBGROUP DECISION · BASELINE PROM GOALS

Clinician _____ Date _____

POST-OP RESTRICTIONS ARE SURGEON-DIRECTED [E] — follow the operating surgeon's protocol for movement, loading, and bracing.

1 · POST-OPERATIVE WOUND, NEUROLOGICAL, DUROTOMY & CES RE-CHECK GATE

MANDATORY EVERY VISIT

- ☐ **Incidental durotomy (n = 1,549, 14 UK centres)** (Tafazal & Sell 2005) [M]: **3.5% primary discectomy; 8.5% stenosis surgery; 13.2% revision discectomy** • screen for postural headache (worse upright, relieved supine) or clear CSF wound leak
- ☐ **Post-op Cauda equina syndrome (CES) & epidural haematoma re-check:** new urinary retention (**Sn 0.90, Sp 0.99**; **PVR > 200 mL → MRI-confirmed CES in 43%**), saddle numbness, or progressive bilateral motor deficit → **same-day emergency** surgical team contact (**decompression < 48 h associated with better outcomes**) [L]

VISIT RE-CHECK: ☐ Wound clean & dry (no infection) ☐ No postural headache / CSF leak ☐ Myotomes & reflexes stable ☐ Bladder / bowel / saddle intact

2 · LUMBAR DISCECTOMY REHABILITATION

OOSTERHUIS COCHRANE + RCT · AFZAL MA

EVIDENCE / PHASE	VERIFIED TRIAL / META-ANALYSIS FINDING & CLINICAL APPLICATION	LEVEL	TICK
Early routine rehab Oosterhuis 2017 J Physiother (RCT)	**no clinically relevant difference vs no referral (function MD 1.5, −3.6 to 6.7); not cost-effective** • focus early care on self-management, walking progression, and neurological vigilance	[M]	☐
Rehab after first-time disc surgery Oosterhuis 2014 Cochrane	**exercise from 4–6 weeks vs no treatment: pain SMD −0.90 (−1.55 to −0.24); function SMD −0.67 (very low / low quality)**	[L]	☐
Active physiotherapy after discectomy Afzal 2022 (MA)	**global improvement OR 1.94** • structured active exercise and functional restoration	[L]	☐
Movement Restrictions	**POST-OP RESTRICTIONS ARE SURGEON-DIRECTED**	[E]	☐

3 · STENOSIS DECOMPRESSION & PRE-OP ACTIVITY

MCGREGOR · SWESPINE

- ☐ **Pre-op activity (Swespine, stenosis surgery)** (Torstensson 2022) [L]: **ODI improvement 19.2 (regular exercise) vs 15.9 (non-regular)**
- ☐ **Rehab after decompression for stenosis (Cochrane)** (McGregor 2013 Cochrane) [M]: **function short-term log SMD −0.22 (−0.44 to 0.00) ≈ 20% improvement; 12 months log SMD −0.26 (−0.46 to −0.05) ≈ 23%**
- ☐ **Restrictions: POST-OP RESTRICTIONS ARE SURGEON-DIRECTED** [E]

4 · LUMBAR SPINAL FUSION REHABILITATION

GREENWOOD MA

- ☐ **Rehab after lumbar fusion (MA)** (Greenwood 2016 Spine) [L]: **"complex rehabilitation" disability ES −0.85 (−1.41 to −0.29) short-term; −0.84 (−1.11 to −0.58) long-term; low quality, n = 237**
- ☐ **Multimodal care:** combine supervised active exercise with cognitive-behavioural management of fear-avoidance [L]
- ☐ **Restrictions: POST-OP RESTRICTIONS ARE SURGEON-DIRECTED** [E]

5 · OPERATIVE PROCEDURE & SURGEON-DIRECTED PROTOCOL LOG

WRITE-IN

Procedure & Level(s) _____ **Surgery Date** _____ **Operating Surgeon** _____ **Durotomy noted?** ☐ No ☐ Yes

Surgeon-Directed Restrictions _____

6 · POST-OP PROGRESSION EVIDENCE

SURGEON-DIRECTED

- ☐ **Disc surgery — active rehab** (Oosterhuis 2014) [M]: no trial reported an increase in reoperation rate; supervised vs home exercise: no differences
- ☐ **Fusion — physiotherapy** (Rushton 2012) [L]: pooled SMD 0.72 (−0.25 to 1.69) at 6 months, 0.52 (−0.45 to 1.49) long term — not significant
- ☐ **Fusion — complex rehab** (Greenwood 2016) [L]: 40 % have significant disability 12 months after fusion; fear avoidance −1.07 and −1.40 (low quality)

7 · CRITERIA TO PROGRESS

E · TICK ALL

- ☐ Surgeon clearance
- ☐ Pain zone OK (p8)
- ☐ Strength LSI on p3
- ☐ Patient confidence
- ☐ Wound healed
- ☐ ROM vs other side
- ☐ Movement quality

[E]

OPERATIVE NOTES · SURGEON-DIRECTED RESTRICTIONS · POST-SURGICAL REHABILITATION PLAN

1 · NICE NG59 — MODALITIES & ORTHOSES NOT RECOMMENDED ("DO NOT OFFER") NICE NG59 GUIDELINE

NICE NG59: Do not offer traction (1.2.6) or passive electrotherapy/orthotic modalities listed below for managing low back pain with or without sciatica.

CATEGORY	NICE NG59 "DO NOT OFFER" RECOMMENDATION & EXACT NUMBER	LEVEL
Spinal Traction	Do not offer traction (1.2.6)	Guideline
Orthoses & Footwear	Do not offer belts or corsets (1.2.3); foot orthotics (1.2.4); rocker-sole shoes (1.2.5)	Guideline
Needling	Do not offer acupuncture (1.2.8)	Guideline
Therapeutic Ultrasound	Do not offer ultrasound (1.2.9)	Guideline
Electrotherapy	Do not offer PENS (1.2.10); TENS (1.2.11); interferential (1.2.12)	Guideline

1B · TRACTION: TIME-LIMITED TRIAL OF AN ADJUNCT NICE NG59 1.2.6 STILL APPLIES

- ☐ **Subgroup RCT** (Fritz 2007) [L]: added traction — Oswestry 7.2 points better at 2 weeks, no difference at 6 weeks; subgroup = leg symptoms + nerve-root signs + peripheralisation with extension or crossed SLR
- ☐ **Subgroup tested (RCT, n = 120)** (Thackeray 2016) [M]: no differences in pain or disability; no subgroup interaction • George 2021 JOSPT CPG: **"Physical therapists should not use mechanical traction for patients with chronic LBP with leg pain"**

Start date _____ Review date _____ ☐ Stop if no change

2 · LUMBAR TRACTION (COCHRANE) WEGNER 2013

- ☐ **Traction (Cochrane)** (Wegner 2013) [M]: **32 RCTs, 2,762 participants; traction alone or with other treatment has little or no effect**

Applies to both mechanical and manual traction in low back pain with or without sciatica; redirect treatment time to active exercise (p8).

3 · THERAPEUTIC ULTRASOUND (COCHRANE) EBADI 2020

- ☐ **Therapeutic ultrasound (Cochrane)** (Ebadi 2020) [L]: **vs placebo, short-term pain MD −7.12 (−17.99 to 3.75) on 0–100 VAS: little or no difference**

No clinically important benefit over placebo; not recommended by NICE NG59 (1.2.9).

4 · SUPERFICIAL HEAT (ACP 2017) & WHO 2023 GUIDELINE GUIDELINES

- ☐ **ACP 2017 (acute/subacute)** (Qaseem 2017) Qaseem 2017: **superficial heat (moderate quality); massage, acupuncture or spinal manipulation (low quality)** — Level: Guideline
- ☐ **WHO 2023 Guideline:** non-surgical management of chronic primary low back pain WHO 2023 — Level: Guideline

Thermal safety screen [E]: check skin integrity and thermal sensation before applying superficial heat; avoid over acute inflammation, impaired sensation, malignancy, or bleeding tissue.

5 · MODALITY & SYMPTOM-MODULATION PARAMETER LOG (BLANK WRITE-IN FIELDS) PARAMETER LOG

RX	MODALITY / ADJUNCT	ANATOMICAL SITE & POSITION	DOSE / PARAMETER	DURATION	RESPONSE & ACTIVE PAIRING
☐	Superficial heat (ACP 2017)				
☐					
☐					
☐					
☐					

PATIENT REASSURANCE · DE-IMPLEMENTATION OF PASSIVE MODALITIES · SELF-MANAGEMENT NOTES

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

1 · SPINAL MANIPULATIVE THERAPY (SMT), MOBILISATION & NICE NG59

NICE NG59 · PACKAGE WITH EXERCISE

- ☐ **NICE manual therapy** NICE NG59 : **"Consider manual therapy (spinal manipulation, mobilisation or soft tissue techniques such as massage) … but only as part of a treatment package that includes exercise" (1.2.7, amended 2026)** — Level: Guideline
- ☐ **SMT for chronic LBP (47 RCTs, n = 9,211)** (Rubinstein 2019 BMJ) M: **similar effect to other recommended therapies for short-term pain: MD −3.17 (−7.85 to 1.51), moderate quality**
- ☐ **Manipulation CPR (see page 4):** Flynn 2002 (**5 criteria; ≥ 4/5 → LR+ 24.38; probability of success 45% → 95%** L) • Childs 2004 (**rule-positive + manipulation: OR 60.8 (5.2–704.7); 92% chance of success; NNT 1.9 (1.4–3.5) at 4 weeks** M)

2 · MASSAGE (COCHRANE — FURLAN 2015)

SHORT-TERM ONLY

- ☐ **Massage (Cochrane)** (Furlan 2015) L: **subacute/chronic vs inactive control, short-term pain SMD −0.75 (−0.90 to −0.60), function SMD −0.72 (−1.05 to −0.39); not in long term**

Per NICE NG59 (1.2.7, amended 2026), deliver soft-tissue massage only as part of a treatment package that includes exercise.

3 · LUMBAR STENOSIS RCT (SCHNEIDER 2019)

NON-SURGICAL RCT

- ☐ **Stenosis — non-surgical (RCT, n = 259)** (Schneider 2019 JAMA Netw Open) M: **medical care vs group exercise vs manual therapy + individual exercise**

Integrates spine/hip mobilisation with individualised conditioning and walking progression.

4 · MANUAL THERAPY & MANDATORY PAIRED EXERCISE LOG

WRITE-IN LOG

RX	TECHNIQUE CATEGORY	TARGET SEGMENT / REGION	IMMEDIATE PRE/POST RESPONSE	PAIRED EXERCISE (NICE NG59 1.2.7)
☐	**Lumbopelvic thrust manipulation (HVLA)**			
☐	**Lumbar PA / rotational mobilisation**			
☐	**Thoracic / hip joint mobilisation adjunct**			
☐	**Soft-tissue massage (Furlan 2015)**			

5 · PRE-MANIPULATION & MANUAL THERAPY SAFETY SCREEN

CONTRAINDICATION GATE

- ☐ **Cauda equina syndrome excluded:** bladder, bowel & saddle sensation intact (p1) L
- ☐ **Vertebral fracture & osteoporosis excluded:** trauma / steroid / multi-flag screen negative (p1) M
- ☐ **Spinal malignancy & infection excluded:** cancer history & systemic fever/infection negative (p1) M
- ☐ **Abdominal aortic aneurysm (AAA) excluded:** no expansile abdominal mass or vascular pain (p1) M
- ☐ **No progressive neurological deficit:** serial myotomes & reflexes stable (p3) E
- ☐ **Anticoagulant / bleeding / post-op fusion hardware screened:** no thrust over unstable/fused segments E

6 · DRY NEEDLING SAFETY & ANATOMICAL DANGER ZONES

SAFETY SCREEN

- ☐ **Adverse event rates:** Brady 2014 M: mild AEs 19.18 %; upper risk rate for significant AEs ≤ 0.04 %; bruising 7.55 %, bleeding 4.65 % • Boyce 2020 M: minor AE 36.7 %; major AE < .1 % (prolonged symptom aggravation .03 %, fainting .02 %, forgotten needles .01 %; major AEs most often with gluteal, lumbar paraspinal, suboccipital and upper trapezius needling)
- ☐ **Serious risks & screening:** Halle 2016 E: pneumothorax, cardiac tamponade, haematoma, infection, CNS injury; filament needles 32–36 gauge (.25–.20 mm) • Kearns 2019 E: anticoagulants / coagulopathy — caution — possible increased incidence or severity of haematoma and bruising
- ☐ **Lumbar / pelvic danger zones:** Valera-Calero 2024 L: piriformis over the sciatic nerve — deeper fascia depth 7.0 ± 1.3 cm • Halle 2016 E: quadratus lumborum — the kidney lies towards the lateral portion of quadratus lumborum

CONSENT OBTAINED FOR: ☐ Spinal manipulation ☐ Spinal mobilisation ☐ Soft-tissue massage ☐ Dry needling | ☐ Risks & alternatives discussed ☐ Part of exercise package (1.2.7) ☐ Patient may decline

Signature _____

MANUAL THERAPY CLINICAL REASONING · IMMEDIATE RE-TEST · EXERCISE INTEGRATION

Exercise Therapy, Secondary Prevention & 12-Week Loading Log

Patient _____

1 · EXERCISE THERAPY FOR CHRONIC LBP (HAYDEN COCHRANE & NMA · NICE NG59) George 2021 CORE INTERVENTION

- ☐ **Exercise therapy, chronic LBP (Cochrane, 249 trials)** (Hayden 2021 Cochrane) [M]: **pain MD −15.2 (−18.3 to −12.2) — clinically important; function MD −6.8 (−8.3 to −5.3) — below MCID**
- ☐ **Exercise types (NMA, 217 RCTs)** (Hayden 2021 J Physiother) [M]: **Pilates, McKenzie, functional restoration: MD −15 to −19 pain, −10 to −12 function vs minimal treatment; "encourage the exercise they enjoy"**
- ☐ **NICE exercise** NICE NG59: **"Consider a group exercise programme (biomechanical, aerobic, mind–body or a combination)" (1.2.2)** — Level: Guideline
- ☐ **Motor control exercise (Cochrane, 29 trials, n = 2,431 — Saragiotto 2016)** [M] & **Pilates (Cochrane — Yamato 2015)** [L]: coordinate lumbopelvic control and progressive movement conditioning tailored to patient preference

2 · COGNITIVE FUNCTIONAL THERAPY (RESTORE) KENT 2023 LANCET

- ☐ **Cognitive functional therapy (RESTORE, n = 492)** (Kent 2023 Lancet) [H]: **RMDQ MD −4.6 (−5.9 to −3.4) at 13 weeks; societal saving −AU$5,276**

☐ Making sense of pain ☐ Exposure with control ☐ Lifestyle coaching

3 · WALKING + EDUCATION (WALKBACK) POCOVI 2024 LANCET

- ☐ **Walking + education (WalkBack, n = 701)** (Pocovi 2024 Lancet) [H]: **recurrence HR 0.72 (0.60–0.85); median days to recurrence 208 vs 112**

Individualised progressive walking and pain education after LBP recovery.

4 · SECONDARY PREVENTION RRS (STEFFENS & SHIRI) PREVENTION

- ☐ **Prevention (exercise + education)** (Steffens 2016 JAMA Intern Med) [M]: **RR 0.55 (0.41–0.74); exercise alone RR 0.65 (0.50–0.86)**
- ☐ **Prevention (exercise alone)** (Shiri 2018 Am J Epidemiol) [M]: **RR 0.67 (0.53–0.85)**

5 · SPONDYLOLYSIS RETURN TO SPORT PARS RTS

- ☐ **Active spondylolysis (RCT):** Selhorst 2026 [H]: **immediate physical therapy: greater early improvement, quicker return to sport and lower LBP recurrence than rest before PT**
- ☐ **RTS timelines:** Panteliadis 2016 [M]: conservative 3.7 months, surgical repair 7.9 months (weighted means) • Grazina 2019 [M]: return to sports at 6.8 months • Bouras 2015 [L]: return to play 6 to 12 months; collision sports prohibited

6 · PAIN-MONITORING MODEL & ACTIVITY MODIFICATION TRAFFIC LIGHT

- ☐ **Pain-monitoring model:** Sprague 2021 [E]: **pain should not exceed 5/10 on the NPRS during or immediately after activity and should return to pre-activity levels by the following morning** • Silbernagel 2007 [M]: continued tendon loading with a pain-monitoring model showed no negative effects
- ☐ **Origin of the zones:** Cavaggion 2024 [E]: low (0–2), acceptable (2–5) and high (5–10) risk zones were based on clinical experience

Low 0–2 (Safe) Acceptable 2–5 (Monitor) High 5–10 (Excessive)

Activity Guidance (E):

AVOID	MODIFY	OK

7 · 12-WEEK EXERCISE & FUNCTIONAL LOADING PROGRESSION LOG 12-WEEK LOG

Primary Programme Ticked: ☐ Directional Preference (McKenzie) ☐ Pilates ☐ Motor Control / Stabilisation ☐ Functional Restoration ☐ CFT Exposure ☐ WalkBack Walking ☐ Group Exercise (NICE 1.2.2)

EXERCISE / FUNCTIONAL TASK	PHASE I (BASELINE)	PHASE II	PHASE III	PHASE IV	WEEK 12 / MAINTENANCE
1 · Primary Exercise / DP					
2 · Trunk / Hip Conditioning					
3 · Functional Lifting / Control					
4 · Walking / Aerobic Volume					
5 · Symptom Response (NPRS)					

EXERCISE PRESCRIPTION DETAILS · SHARED DECISION-MAKING ("ENCOURAGE THE EXERCISE THEY ENJOY") · PREVENTION PLAN

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____

#	DATE	NPRS	DP / CENTR.	NEURO / SLR	ROM	EXERCISE & CFT DELIVERED	MANUAL & ADJUNCTS	PROGRESSION	ADHERENCE	PATIENT SIGNATURE	INIT
1						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
2						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
3						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
4						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
5						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
6						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
7						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
8						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
9						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
10						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
11						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			
12						DP | Pilates | MCE | Func | CFT | Walk | Group	SMT | Mob | Massage | Heat | Neuro	Low | Mod | High			

DISCHARGE & OUTCOME MILESTONE SUMMARY (CHANGE VS VERIFIED MCID)

FINAL OUTCOME

MIC consensus (Ostelo 2008) [M]**:** VAS 15; NRS 2; RMDQ 5; ODI 10; or 30% improvement from baseline.

LUMBAR PROM & VERIFIED THRESHOLD	INITIAL	DISCHARGE	Δ	MCID MET
ODI (MIC 10 or 30% • Mod. ODI MCID ≈ 6 points)				☐
RMDQ (MIC 5 or 30% reduction + global rating)				☐
NPRS (MDC 2; MCID 2.2 at 1 week / 1.5 at 4 weeks; NRS 2)				☐
PSFS (MCID 2 points ROC; 1.4 SEM method)				☐
TSK-11 (reduction ≥ 4 points = important change)				☐
STarT Back (0–3 low; items 5–9 4–5 high) / ÖMPSQ-10 (cut-off 50)				☐

Discharge: ☐ Goals & MCID met ☐ WalkBack / self-management plan ☐ Specialist / surgical referral ☐ Non-responder

OBJECTIVE RE-TEST MILESTONES

SERIAL RE-TEST

OBJECTIVE MEASURE	BASE	MID	D/C	MET
Lumbar Flexion / Extension				☐
Centralisation / DP Status				☐
Myotome / Reflex / Sensory				☐
Neurodynamics (SLR / Slump / FST)				☐
McGill Endurance Ratios				☐
Aberrant Movements Resolved				☐

Session 6 checkpoint: Re-score the main PROM against the MCID / MDC already on the sheet. If there is no change, re-assess the diagnosis, the dose or referral (E).

HOME PROGRAMME · WALKBACK PREVENTION PLAN · DISCHARGE SUMMARY

Physician follow-up / sign-off _____ **Signature** _____ **Date** _____

Discharging Clinician _____ **Sig.** _____